Meftahul Jannat

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Research Interests

Machine Learning, Computer Vision, LLM, Bioinformatics, and Cyber Security

Education

B.Sc. in Electronics & Telecommunication Engineering 2020 – 2025
Rajshahi university of Engineering & Technology *(RUET)*
Result: CGPA 3.59/4.00
Position: 7th
Thesis: Lung Segmentation with Lightweight Convolutional Attention Residual U-Net
Algorithm: CNN, Residual U-Net

Higher Secondary Certificate 2018 – 2019
Rajshahi College, Rajshahi
Result: GPA 5.00/5.00

Secondary School Certificate 2012 – 2017
Govt. Laboratory High School, Rajshahi
Result: GPA 5.00/5.00

Experience

Lecturer, Dept. of ICT, Jan 2025 – Current
Bangladesh Army University of Science and Technology (BAUST), Saidpur

- Delivered lectures and interactive coursework across core Information & Communication Technology domains, including Analog Communication & Radio-TV Engineering, Optical Fiber Communication, and Wireless & Mobile Communication, Data Communication & Computer Networks and Management Information Systems (MIS)
- Supervised undergraduate final-year Capstone Projects, guiding students through problem formulation, theoretical modeling, simulation, hardware prototyping, and final thesis defense preparation.

Trainee Engineer, IC Physical Design (PnR) – Ulkasemi Pvt. Limited Dec 2025 – Jan 2026

- Execute Place and Route (PnR) flows under supervision, translating synthesized gate-level netlists into physical chip layouts using industry-standard EDA tools such as Cadence Innovus, Synopsys IC Compiler.
- Develop automation scripts utilizing Tcl, Python, and Bash/Perl to streamline physical design tasks, report parsing, and EDA tool constraint management.

Research Assistant, AIMS Lab – United International University Nov 2025 – Nov 2025

- Conduct research on automated Scribe technology AIMScribe at AIMS Lab (UIU), architecting novel pipelines that integrate Speech-to-Text (STT), Pyannote speaker diarization, Vision-Language Models (VLMs), and LLMs for multimodal document generation.
- Evaluated and experimented deep learning models across speech, vision, and NLP modalities, utilizing rigorous cross-validation and statistical evaluation metrics (WER, DER, BLEU/ROUGE scores) to validate model performance.

Publications

Real-time jute leaf disease classification using an explainable lightweight CNN via a supervised. and semi-supervised self-training approach Oct 2025
Meftahul Jannat, Md Shahab Uddin, Mohammad Asif Hasan, Md Saimun Alam,Avijit Paul,
Muhammad EH Chowdhury, Julfikar Haider
Frontiers in Plant Science **–** DOI: 10.3389/fpls.2025.1647177

Lung Segmentation with Lightweight Convolutional Attention Residual U-Net. March 2025
Meftahul Jannat, Shaikh Afnan Birahim, Mohammad Asif Hasan, Tonmoy Roy, Lubna Sultana, Hasan Sarker, Samia Fairuz, Hanaa A Abdallah
Diagnostics – DOI: 10.3390/diagnostics15070854

Cotton Leaf Disease (CLD) Classification Using Multihead External Attention Vision Transformer. Dec 2024
Meftahul Jannat, Hasan Sarker
27th International Conference on Computer and Information Technology (ICCIT)
DOI: 10.1109/ICCIT64611.2024.11022519

Projects

CXR-Agent: Vision-Language Model for Chest X-Ray Analysis. github.com/meftahuljannat19/cxr-agent

- Finetune Vision LLM to generate medical reports from Chest X-ray images using Unsloth
- Finetuned different LLMs (Llama-3.2-11B-Vision-Instruct, Qwen2-VL-7B-Instruct) on a publicly available chest x-ray dataset (Chexpert, MIMIC)

Automated Toll Collection System using number plate recognition with YOLOv8 and EasyOCR on custom dataset annotated with CVAT github.com/meftahuljannat19/Automated_Toll_System_YOLOv8

- Annotated a custom vehicle dataset of total 350 vehicle (car, bus, truck) using CVAT then trained YOLOv8 model on this custom dataset to detect different vehicles.
- The number plate is detected using edge detection and recognized with EasyOCR, enabling an automated toll payment system. The toll gate is then automatically controlled through a microcontroller, ensuring a seamless and efficient process.

Technologies

Languages: C++, C, Python, MATLAB, VHDL
Framework: PyTorch, Tensorflow, Keras, Unsloth
AI/ML Techniques: CNN, LSTM, NLP, Transformers, YOLO, LLM, Explainable AI, GAN
Development Tools: FastAPI
Hardware: Arduino, Raspberry Pi, ESP32
Software: MATLAB, COMSOL Multiphysics

Awards & Achievements

Research Publication Award June 2025
RUET INNOVISTA 2025

References

Dr. Md. Kamal Hossain
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